

COAGULANTS AND ANTICOAGULANTS

12/2/2025

Learning Outcomes

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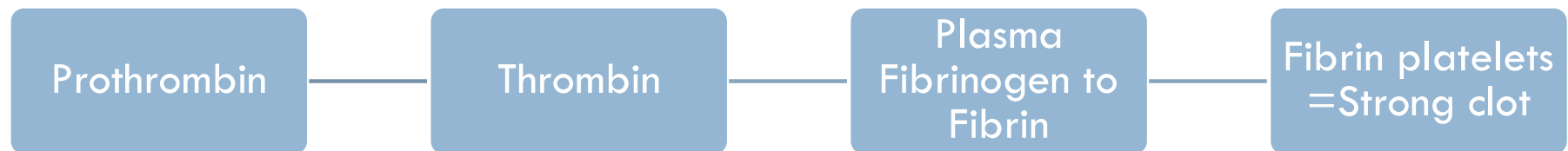
Upon completion of this lecture a student will be able to

- **Describe hemostasis**
- **Enumerate pro-coagulant agents**
- **Explain the therapeutic uses of coagulants in dentistry**
- **List types of anticoagulants and antiplatelets**
- **Describe the mechanism of action of different anticoagulants**

INTRODUCTION TO DISORDERS OF COAGULATION

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- Hemostasis refers to the finely regulated dynamic process of maintaining fluidity of the blood, repairing vascular injury, and limiting blood loss while avoiding vessel occlusion (thrombosis) and inadequate perfusion of vital organs.
- Either extreme—excessive bleeding or thrombosis—represents a breakdown of the hemostatic mechanism.



Common causes of dysregulated hemostasis include hereditary or acquired defects in the clotting mechanism and secondary effects of infection or cancer.

Adhesion and activation of platelets and fibrin formation.

Platelet activation leads to formation of haemostatic plug, which stops bleeding and reinforced by fibrin.

A BLOOD CLOT

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Consists of
platelets meshed
into fibrin



A web-like
accumulation of
strands with RBCs



There are two major facets of the
clotting mechanism – the platelets,
and the thrombin system

Coagulation Factors

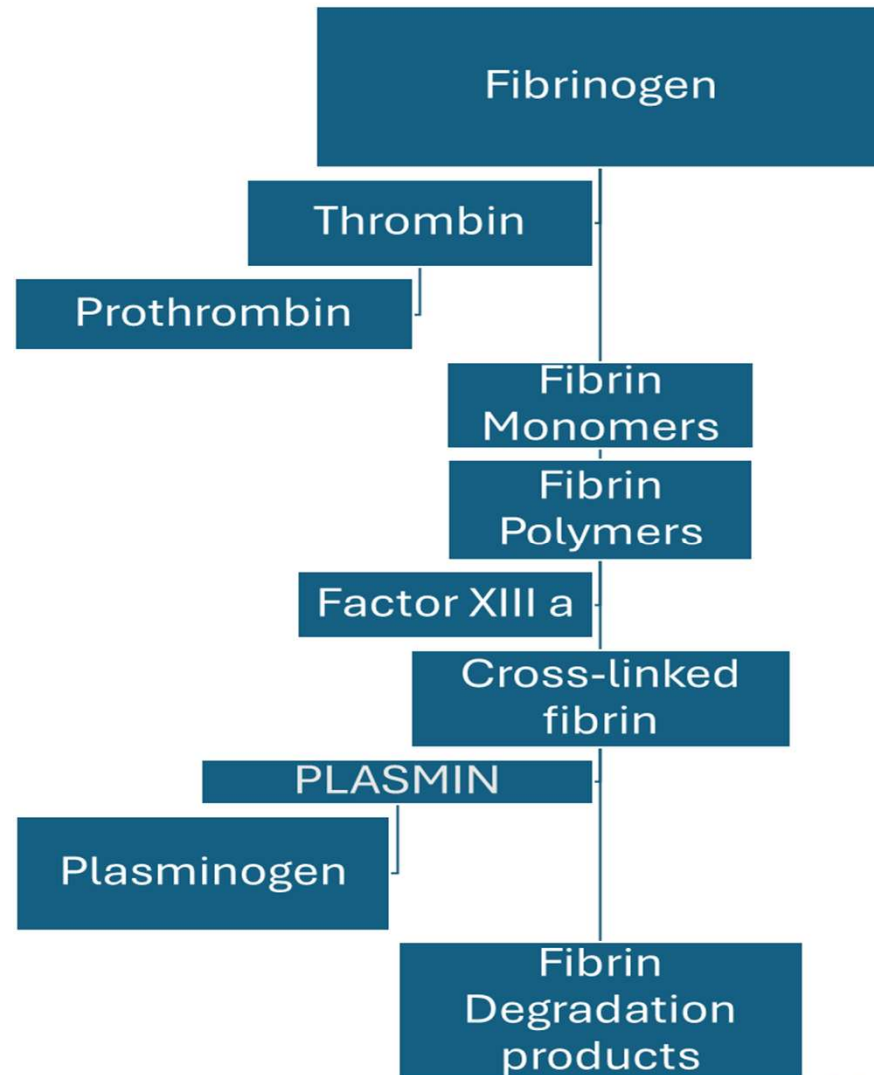
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Factor	Name
I	Fibrinogen
II	Prothrombin
III	Tissue Factor or thromboplastin
IV	Ca ⁺⁺
V	Proaccelerin
VII	Proconvertin
VIII	Antihemophilic A factor
IX	Antihemophilic B factor or Christmas factor

Factor	Name
X	Stuart or Stuart-Prower factor
XI	Plasma thromboplastin antecedent
XII	Hageman factor, contact factor
XIII	Fibrin stabilizing factor Prekallikrein factor High-molecular-weight kininogen

Coagulation Cascade

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COAGULANTS

Coagulants

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Definition: Agents that promote blood clotting.

Types

Topical Coagulants:

- **Examples:** Fibrin glue, Thrombin, Gelatin sponges.
- **Applications:** Post-surgical bleeding in dentistry.

Systemic Coagulants:

- **Vitamin K:** Role in the synthesis of clotting factors.
- **Tranexamic Acid:** Anti-fibrinolytic agent.

Vitamin K



1. Natural form: Vit.K₁ and K₂ are fat-soluble. K₁ is found in food and K₂ is synthesized by intestinal bacteria(Ex: E.coli), both require bile salts for absorption from the intestinal tract.



2. Synthetic form: Vit K₃ and K₄ are water-soluble.



Dose- 0.5 to 1.0 milligrams given intramuscularly.



Mechanism: Vitamin K is necessary for the final stage of synthesis of coagulation factors (mainly factor II, VII, IX and X) in the liver.

Uses

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Therapeutically, vitamin K is used in prophylaxis and treatment of deficiency of clotting factor due to:

- Dietary deficiency of vitamin K
- Chronic antimicrobial therapy
- Malabsorption syndrome
- Jaundice, liver diseases such as cirrhosis and hepatitis
- To counteract the overdosing of oral anticoagulants

VITAMIN K ANALOGUES: Menadione

TRANEXAMIC ACID

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Tranexamic acid produces an antifibrinolytic effect

It is widely distributed throughout the body and has less protein binding. It diffuses across the placenta and is distributed into breast milk.

Adverse effects include dose-related gastrointestinal disturbances, hypotension particularly after rapid IV administration and vision disturbances rarely.

It is used for prophylaxis and control of bleeding after dental extraction.

Other agents

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Specific Clotting Factors. Patients that have an acquired or hereditary clotting factor deficiency.



Desmopressin Acetate. Desmopresin has the unique activity of producing a dose-related increase in circulating Factors.



Botropase and Ethamsylate: primary and secondary postoperative internal and external hemorrhage.



The use of adsorbent chemicals, such as zeolites, and other hemostatic agents are used for sealing severe injuries quickly.

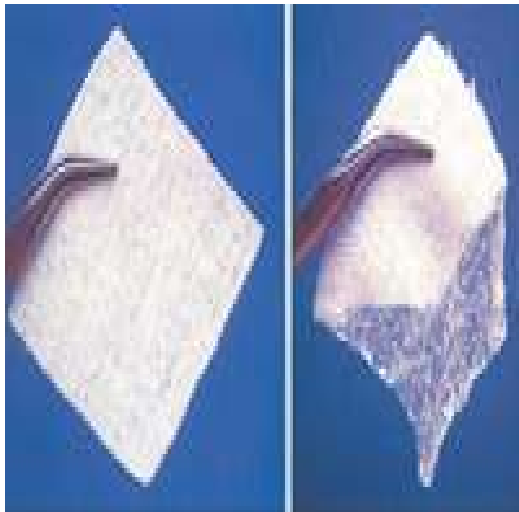
Overall uses

Bleeding caused by lack of Vitamin K

Overdose of oral anticoagulants

Long use of broad spectrum antibiotics

Hemorrhage



TOPICAL HEMOSTATIC AGENTS

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Hemostatic Collagen

These products are fabricated from bovine collagen and are nontoxic and nonpyrogenic. The products are highly absorbent.



Gelatin

one of the more commonly employed agents for the control of minor bleeding. It is a porous, sponge.



Bone Wax

Bone Wax (Ethicon) is a sterile mixture of beeswax, paraffin, and isopropyl palmitate.



Cellulose

Resorbable cellulose material

What should we do ??

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If an intraoperative bleeding episode is encountered, the clinician should consider several steps.

A quick mental review of the patient's medical history is first.

injection of epinephrine into the area may be needed. This will likely provide a temporary reduction of bleeding as a result of local vasoconstriction.

The process of immediate delivery of the patient to a medical facility for possible transfusions, anticoagulant reversal, and general life support measures can be initiated.

However, a careful clinician will rarely encounter such an event.

Anticoagulant Drugs

Anticoagulants

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Definition: Agents that prevent blood clot formation.



Types:

Oral Anticoagulants:

- **Warfarin:** Vitamin K antagonist.
- **Direct oral anticoagulants :** Dabigatran, Rivaroxaban.

Parenteral Anticoagulants:

- **Heparin:** Activates antithrombin III.
- **Low-molecular-weight heparins (e.g., Enoxaparin).**

Anticoagulant Use

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Anticoagulant drugs help prevent the development of harmful clots in the blood vessels by lessening the blood's ability to cluster together.



Heparin and warfarin are the two traditional anticoagulants.



Anticoagulants are used for acute coronary syndromes, deep-vein thrombosis (DVT), pulmonary embolism (PE), and heart surgery.

Heparin

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Heparin is a naturally occurring anticoagulant produced by basophils and mast cells to prevent the formation and extension of blood clots. It inhibits clotting factors.



Low-molecular weight heparin is more convenient and cost-effective. It has similar results as heparin.



Administered by subcutaneous injection.

Warfarin

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Warfarin is an oral medication.

It decreases blood coagulation by interfering with vitamin K metabolism.

It stops the blood from clotting within the blood vessels and is used to stop existing clots from getting bigger and to stop parts of clots from breaking off and forming emboli.

The most common side effect of **warfarin** is bleeding. **Warfarin** inhibits the synthesis of clotting factors: II, VII, IX and X.

Dicumurol

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It is a potent oral anticoagulant that acts by inhibiting the synthesis of vitamin K-dependent clotting factors (prothrombin and factors VII, IX and X); it is starting to largely replace warfarin.



It is used especially in preventing and treating thromboembolic disease.

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FIBRINOLYTICS

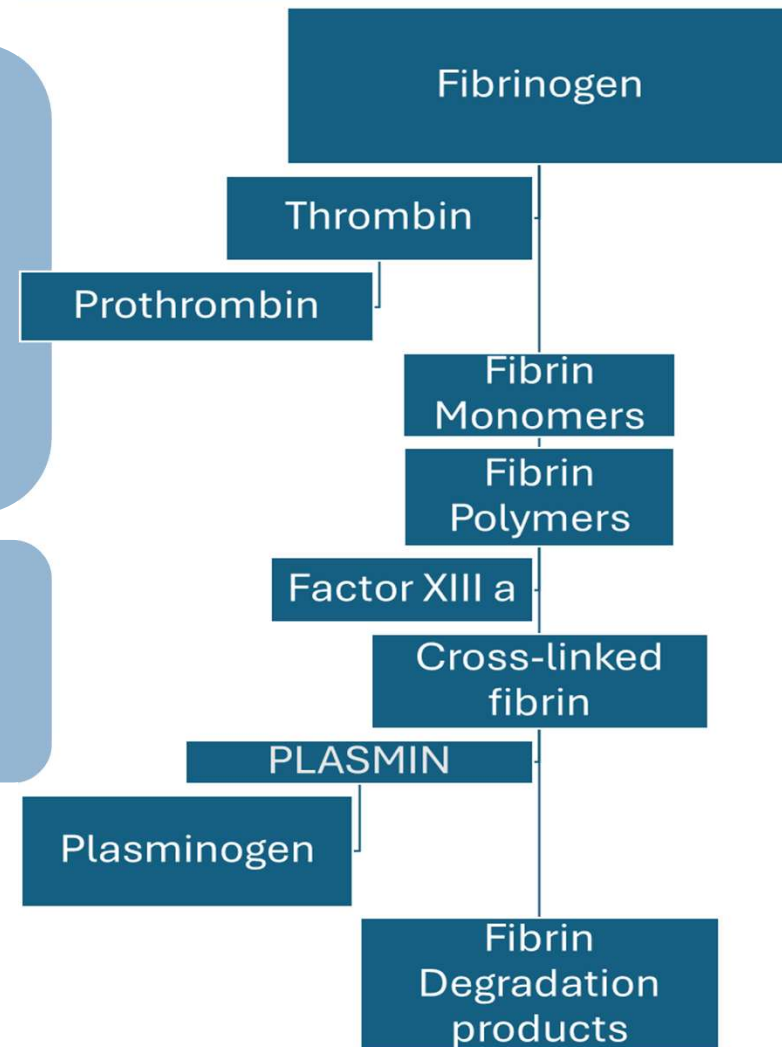
FIBRINOLYTICS

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These are agents used to lyse clots to recanalize occluded blood vessels, mainly used in coronary arteries.

STREPTOKINASE

- It converts **plasminogen to plasmin**.
- It is given by the parenteral route and has a short plasma half-life.



PLATELET INHIBITING DRUGS

PLATELET INHIBITING DRUGS

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ASPIRIN: In small doses, aspirin inactivates irreversibly platelet **enzyme cyclooxygenase**.



The effect of enzyme inactivation lasts till the life of the platelet.



It is indicated in prophylaxis in cases of increased risk of blood clotting, myocardial infarction, coronary bypass, and ischemic attack.

Dental Considerations

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For patients on
Anticoagulants:

Risk of bleeding during and after procedures.

Monitoring for patients on Warfarin.
Adjusting anticoagulant therapy pre-procedure (if required).



For **Coagulants:**

Complications like clot migration.

Management of Bleeding in Dental Practice

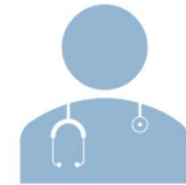
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Local measures: Pressure application, sutures, and topical agents.



Use of **Tranexamic acid** mouthwash.



Consultation with a physician for systemic management.

Books to Read

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Bart –Jhonson, Frank J. Dowd. Pharmacology and Therapeutic for Dentistry, 6th edition, 2011. Elsevier Publishers,USA



Karen Whalen, Richard Finkel, Thomas A Panavelil. Lippincott Illustrated Reviews Pharmacology. 6th ed. 2015. Philadelphia Wolter Kluwer Puplisher.



Katzung, Bertram, and Anthony Trevor. Basic and Clinical Pharmacology. 13th ed, 2015. New York: Lange Publishers.